SIMATS ENGINEERING

CSA0705 - COMPUTER NETWORKS

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1. Introduction

In a modern factory, Automated Guided Vehicles (AGVs) move materials along fixed routes. An AGV must receive commands over a wireless network while it is moving. As it moves, it must keep switching from one access point (or cell) to the next. This switch is called a **handover**, and communication may pause during it.

This report compares **Wi-Fi** and **private 5G** for AGV communication using the figures given in the assignment. It also explains the three 5G service types, the role of Bluetooth, Zigbee and LoRa in a plant, Wi-Fi 6 with fast roaming (IEEE 802.11r), and whether LoRaWAN can carry the traffic of 600 sensors.

2. Given Data and Assumptions

Parameter	Value
AGV speed	2 m/s
Route length	1.2 km = 1200 m
Wi-Fi access-point spacing	30 m
Wi-Fi handover time	approximately 150 ms
Private 5G cell coverage	300 m
5G handover time	approximately 20 ms
AGV command deadline	10 ms
Number of sensors (Part E)	600
Data per sensor (Part E)	50 bytes/second
LoRa gateway capacity (Part E)	approximately 5 kbps

Assumptions:

- The handover count is a **simplified interval estimate**: route length divided by access-point spacing (or cell coverage). It is not a measured value.
- The AGV moves at a constant speed of 2 m/s, and 1 byte = 8 bits.
- No new performance figures have been added. Only the values in Table 1 are used.

3. Part A – Handover Calculations

3.1 Journey time

Journey time = Distance ÷ Speed
= 1200 m ÷ 2 m/s = 600 s = 10 minutes

Journey time = 600 seconds (10 minutes)

3.2 Wi-Fi handovers and interruption

Handovers = Route length ÷ AP spacing = 1200 m ÷ 30 m = 40
Total interruption = 40 × 150 ms = 6000 ms = 6 s

Wi-Fi: 40 handovers, 6 seconds not reachable per trip

3.3 Private 5G handovers and interruption

Handovers = Route length ÷ Cell coverage = 1200 m ÷ 300 m = 4
Total interruption = 4 × 20 ms = 80 ms = 0.08 s

Private 5G: 4 handovers, 0.08 seconds not reachable per trip

3.4 Consistency check

The time between handovers is the spacing divided by the speed. Dividing the journey time by this gives the same counts:

Wi-Fi: 30 m ÷ 2 m/s = 15 s between handovers; 600 s ÷ 15 s = 40
5G: 300 m ÷ 2 m/s = 150 s between handovers; 600 s ÷ 150 s = 4

3.5 Comparison of Wi-Fi and 5G

Item	Wi-Fi	Private 5G
Handover count per trip	40	4
Time per handover	150 ms	20 ms
Total time not reachable	6000 ms = 6 s	80 ms = 0.08 s
Share of the 600 s journey	6 ÷ 600 = 1.00 %	0.08 ÷ 600 = 0.013 %
Time per handover compared with 10 ms deadline	150 ÷ 10 = 15 times longer	20 ÷ 10 = 2 times longer

Interruption ratio = 6000 ms ÷ 80 ms = 75

Discussion: In this example, Wi-Fi's total interruption is **75 times** that of private 5G, because 5G needs ten times fewer handovers (4 against 40) and each handover is shorter (20 ms against 150 ms).

Important notes:

- These are **simplified interval-based estimates**. The exact number of handovers depends on the starting position, the ending position, and the actual placement of access points or cells.
- The 5G handover time of 20 ms is **still greater than the 10 ms command deadline**. So private 5G has a much shorter interruption, but the given figures alone do **not** guarantee that every command will meet its deadline.

4. Part B – 5G Service Types

Service	Full name	Main feature
eMBB	Enhanced Mobile Broadband	High data rates and high capacity, for large amounts of data such as video.
URLLC	Ultra-Reliable Low-Latency Communications	Very reliable delivery with low delay, for important messages that must arrive quickly.
mMTC	Massive Machine-Type Communications	Supports very large numbers of connected devices that send small amounts of data.

4.1 Mapping of equipment to service

Equipment	Recommended service	Reason
AGVs	**URLLC**	Requires reliable, low-latency communication for movement and control.
Cameras	**eMBB**	Requires high bandwidth for video transmission.
Sensors	**mMTC**	Supports a large number of connected devices with small data transmissions.

- **AGVs → URLLC:** a late or lost control command can stop or misdirect a vehicle, so reliability and low delay matter more than speed.
- **Cameras → eMBB:** video produces a continuous stream of large data, so high bandwidth is the main need.
- **Sensors → mMTC:** a plant may have hundreds of sensors, each sending only small readings, so supporting many devices is more important than speed.

URLLC is designed for low delay, but this report does not claim any measured latency. The 5G handover time given in the assignment (20 ms) is still above the 10 ms deadline.

5. Part C – Bluetooth, Zigbee and LoRa

Some plant tasks do not need the high speed of Wi-Fi or 5G. For them, simpler, low-power technologies can be a better choice.

Bluetooth is a short-range technology. It is better than Wi-Fi or 5G for maintenance tools and nearby peripherals, such as a technician's handheld device, where a simple, low-power, close-range link is enough.

Zigbee is a low-power technology for small sensors and monitoring devices. It suits small sensor networks in a local area where devices send small messages and should run for a long time on little power.

LoRa / LoRaWAN gives long-range communication for low-data-rate traffic. It suits remote and battery-powered sensors that send small readings occasionally, over distances where Wi-Fi coverage would be costly to install.

5.1 Why they are not suitable for CCTV or time-critical AGV control

- **High-bandwidth CCTV** sends a continuous stream of large video data. Bluetooth, Zigbee and LoRaWAN are designed for small, low-rate data, so they cannot carry it. Wi-Fi or 5G (eMBB) is needed.
- **Time-critical AGV control** needs reliable commands within a short deadline (10 ms in this assignment). These low-power technologies are not designed for continuous, fast, dependable delivery to a moving vehicle.
- Their low power and low data rate are their strengths for sensors, but the same features make them unsuitable for fast, heavy or critical traffic.

5.2 Comparison table

Technology	Range	Data rate	Power use	Best use in a plant	Suitable for CCTV / AGV control?
Bluetooth	Short	Low to moderate	Low	Maintenance tools, nearby peripherals	No
Zigbee	Short to medium (local network)	Low	Very low	Small sensor networks and monitoring	No
LoRaWAN	Long	Very low	Very low (battery sensors)	Remote, low-rate, non-critical sensors	No
Wi-Fi	Medium (access-point coverage)	High	Higher	General plant networking, cameras	CCTV: yes. AGV: needs testing
Private 5G	Wide (cell coverage)	High	Higher	Mobile equipment, cameras, many devices	CCTV: yes (eMBB). AGV: preferred, needs validation

Table 5: Comparison of five wireless technologies (qualitative).

6. Part D – Wi-Fi 6 and Fast Roaming (802.11r)

6.1 What is Wi-Fi 6?

Wi-Fi 6 (IEEE 802.11ax) is the sixth generation of Wi-Fi. It is designed to work better when many devices share the network, giving better efficiency and capacity than earlier Wi-Fi versions. It is still Wi-Fi, so a moving device must still roam from one access point to another.

6.2 What does IEEE 802.11r mean?

IEEE 802.11r is a Wi-Fi standard amendment called **Fast BSS Transition**, or **fast roaming**. A BSS (Basic Service Set) is the area served by one access point.

6.3 How does fast roaming reduce delay?

Without it, a device that moves to a new access point has to repeat the security authentication and key set-up with that access point, which takes time. With 802.11r, the security keys are prepared in advance, so the device can join the next access point with fewer message exchanges. This reduces the authentication part of the handover delay. Other parts, such as finding the next access point, still take time.

6.4 Can it meet the 10 ms deadline?

The assignment gives a Wi-Fi handover time of 150 ms. No separate handover time is given for Wi-Fi 6 with 802.11r, and none is invented here.

Handover time ÷ Deadline = 150 ms ÷ 10 ms = 15

The given handover time is **15 times** the command deadline. Fast roaming may reduce the roaming delay, but the given information does **not** prove that it can bring the delay below 10 ms. Only practical measurement in the actual plant can show this.

6.5 Remaining risks

Risk	Why it matters for AGV commands
Interference	Other radio signals and machinery can disturb the wireless link.
Packet loss	Lost commands must be resent, causing delay.
Network congestion	Many devices sharing the network can delay commands.
Jitter	Varying delay means some commands may arrive after the deadline even if the average delay is small.
Handover interruption	The AGV may be unreachable while switching access points.

Conclusion for Part D: fast roaming can reduce roaming delay, but actual performance must be measured and verified before Wi-Fi 6 is trusted for a 10 ms AGV command deadline.

7. Part E – LoRaWAN Capacity Calculation (Alternative Concept Analysis)

The proposal is to use LoRaWAN for sensors instead of Wi-Fi. Given: 600 sensors, 50 bytes/second per sensor, and an approximate gateway capacity of 5 kbps.

Step 1 – Convert bytes to bits

Data per sensor = 50 bytes/s × 8 bits/byte = 400 bits/s

Step 2 – Total data from 600 sensors

Total = 600 × 400 bits/s = 240,000 bits/s = 240 kbps

Step 3 – Compare with gateway capacity

Required ÷ Available = 240 kbps ÷ 5 kbps = 48
The sensor traffic is 48 times the stated gateway capacity

Step 4 – Maximum number of sensors under the stated capacity

Maximum sensors = 5000 bits/s ÷ 400 bits/s per sensor = 12.5
About 12 sensors (whole sensors) in an ideal calculation

This ideal count ignores protocol overhead and other limitations, so the real number would be lower.

Technical note

The value of 5 kbps is treated here as the **approximate aggregate capacity supplied by the assignment**. It is not claimed that every real LoRaWAN gateway has exactly this capacity. Real LoRaWAN capacity depends on factors such as the spreading factor, airtime, regional duty-cycle limits, protocol overhead and collisions.

Conclusion and recommendation

One gateway with the stated capacity **cannot support all 600 sensors** sending 50 bytes/second each. Recommended use:

- Use LoRaWAN for a **small number of remote sensors** that send small, infrequent, non-critical readings.
- Use Wi-Fi or wired connections for **high-data-rate sensors** and devices that need frequent or time-sensitive communication.
- Do **not** use LoRaWAN for high-bandwidth video or real-time AGV control.

8. Overall Comparison Table

Application	Recommended technology	Reason	Caution
AGV control	Private 5G (URLLC)	4 handovers and 0.08 s total interruption, against 40 handovers and 6 s for Wi-Fi	20 ms handover is above the 10 ms deadline; must be validated
AGV control (alternative)	Wi-Fi 6 with 802.11r	Fast roaming can reduce roaming delay	Given 150 ms is 15 times the deadline; must be tested
Cameras / CCTV	5G eMBB (or Wi-Fi)	High bandwidth for video	Not for Bluetooth, Zigbee or LoRaWAN
Large numbers of sensors	5G mMTC	Many devices with small data	High-rate sensors may need Wi-Fi or wired links

Application	Recommended technology	Reason	Caution
Few remote, low-rate sensors	LoRaWAN	Long range, low power	600 sensors need 48 times the stated capacity; use a subset only
Small local sensor network	Zigbee	Low power, local monitoring	Low data rate only
Maintenance tools, peripherals	Bluetooth	Short-range, simple links	Short range only

Table 7: Overall technology selection.

9. Final Recommendation

- **AGVs:** private 5G is preferable because it gives fewer handovers (4 against 40) and lower total interruption (0.08 s against 6 s) in the given scenario.
- However, its stated 20 ms handover time is **still greater than the 10 ms command deadline**, so it must be validated before use.
- **Wi-Fi 6 with 802.11r** may reduce roaming delay, but it needs practical testing before it can be trusted for this deadline.
- **Service types:** eMBB for cameras, URLLC for AGVs, and mMTC for large numbers of sensors.
- **Bluetooth, Zigbee and LoRaWAN** are useful for specific low-power or short-range plant applications.
- **LoRaWAN** should be used only for a suitable subset of low-rate sensors, because the calculated traffic of all 600 sensors (240 kbps) exceeds the given gateway capacity (5 kbps).

10. Conclusion

For the AGV route of 1200 m at 2 m/s, a Wi-Fi network would cause 40 handovers and 6 seconds of unreachable time per 10-minute trip, while private 5G would cause 4 handovers and 0.08 seconds. Private 5G is therefore the better choice on the given figures, but its 20 ms handover is still above the 10 ms deadline, and Wi-Fi 6 with 802.11r also needs measurement. Different plant devices need different technologies: URLLC for AGVs, eMBB for cameras, mMTC for many sensors, and Bluetooth, Zigbee or LoRaWAN for suitable low-power or short-range tasks. **A wireless network should be chosen by reliability, delay, mobility and capacity, not by speed alone.**